

CSA0609-DESIGN AND ANALYSIS OF ALGORITHMS

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Exp-08: Insertion Sort with Duplicate Elements

Aim:

To sort an array containing duplicate elements using Insertion Sort and analyze its time complexity.

Algorithm:

1. Read the size of the array.
2. Read the array elements, including duplicate values.
3. Consider the second element as the first key.
4. Compare the key with elements in the sorted portion.
5. Shift larger elements one position to the right.
6. Insert the key at its correct position.
7. Repeat until all elements are processed.
8. Display the sorted array and complexity.

Source Code:

```
n = int(input("Enter number of elements: "))
arr = list(map(int, input("Enter array elements: ").split()))

for i in range(1, n):
    key = arr[i]
    j = i - 1

    while j >= 0 and arr[j] > key:
        arr[j + 1] = arr[j]
```

```
j -= 1
```

```
arr[j + 1] = key
```

```
print("Sorted Array:", arr)
```

```
print("Time Complexity: O(n^2)")
```

```
print("Space Complexity: O(1)")
```

Output:

```
Enter number of elements: 7
Enter array elements: 5 2 8 2 3 5 1
Sorted Array: [1, 2, 2, 3, 5, 5, 8]
Time Complexity: O(n^2)
Space Complexity: O(1)
```

Result:

The array containing duplicate elements was successfully sorted using Insertion Sort.